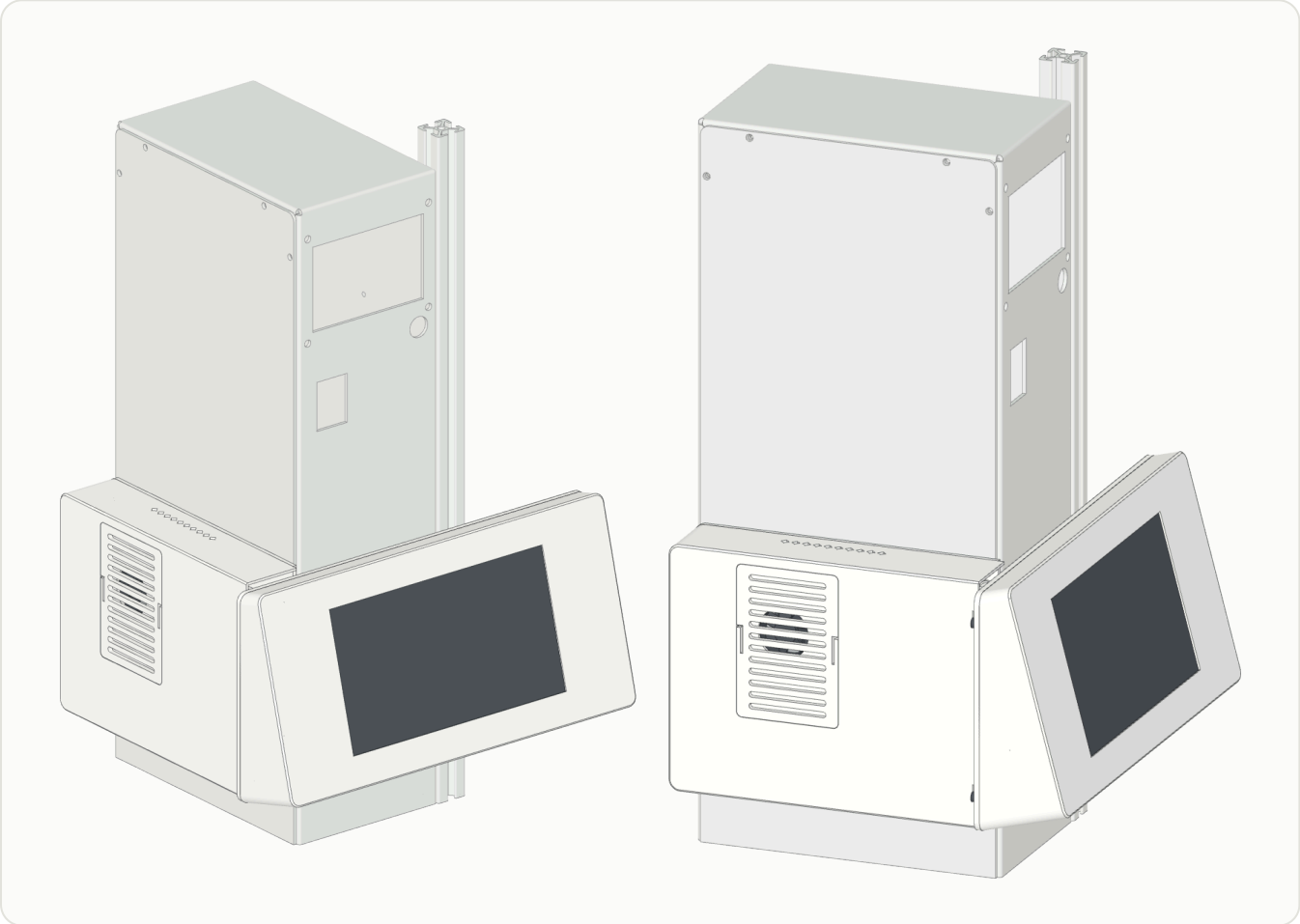
PRINTDECK

LulzBot TAZ Workhorse mount

A mount for the Printdeck at the front left of the TAZ Workhorse: the 7 inch touch display in a wide head below the TAZ Workhorse logo, and the Raspberry Pi 5 in a pod on the left side of the electronics chassis. The head leans back 15°, the same as the Slate, and faces the front of the printer. The drive is in the head, behind the display. There is no hole to drill: two printed clips hold on the front rail with metal T-nuts, the upper pair of screws of the cover (PP-FP0164) holds the plate, and the cables leave at the rear of the chassis.



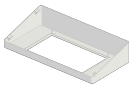
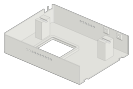
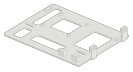
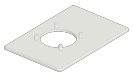
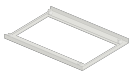
Navy PLA · Orange PLA · White PLA

8 printed parts · 344 g · 16 h 42 min · No supports

No drilling in the printer

Print plan

Print the parts before you start. The numbers come from slices of the STL files with the Cura profiles of this mount (CuraEngine, LulzBot TAZ Workhorse SE, 0.5 mm nozzle, 2.85 mm PLA), with the skirt.

	File	Colour	Size on the bed (mm)	Filament	Length	Time	Cost
	`stl/head.stl`	Navy	252 × 126 × 50	87 g	11.0 m	4 h 28 min	$2.00
	`stl/pod.stl`	Navy	173 × 135 × 36	95 g	12.1 m	4 h 42 min	$2.18
	`stl/back.stl`	Orange	252 × 131 × 10	69 g	8.7 m	3 h 16 min	$1.59
	`stl/plate.stl`	Orange	173 × 135 × 23	61 g	7.7 m	2 h 18 min	$1.40
	`stl/clip.stl` (2×)	Orange	6 × 16 × 24	8 g	0.8 m	0 h 36 min	$0.18
	`stl/grille.stl`	White	58 × 84 × 3	12 g	1.5 m	0 h 42 min	$0.28
	`stl/fan_plate.stl`	White	57 × 84 × 6	12 g	1.5 m	0 h 40 min	$0.28
	Total			**344 g**	**43 m**	**16 h 42 min**	**$7.91**
	*`stl/fit_display.stl`*	*Navy*	*optional*	*22 g*	*2.7 m*	*1 h 10 min*	*$0.51*

By colour

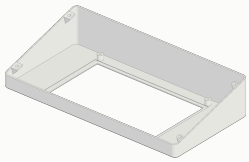
Colour	Parts	Filament	Length	Time	Spools
Navy	head, pod	182 g	23.1 m	9 h 10 min	1 × 1 kg (18 %)
Orange	back, plate, 2 × clip	138 g	17.2 m	6 h 10 min	1 × 1 kg (14 %)
White	grille, fan_plate	24 g	3.0 m	1 h 22 min	1 × 1 kg (2 %)

Total. 344 g of PLA (43 m), 16 h 42 min of printing, $7.91 at $22.99/kg. With the fit test: 366 g and 17 h 52 min. The largest part (the head) needs a bed of 252 × 126 mm; the Workhorse bed is 280 × 280 mm. Until you have the metal T-nuts: 2 printed roll-in nuts (`stl/roll_nut.stl`), 2 g and 0 h 06 min.

Order. Print the fit test first (1 h 10 min): the display pocket has 0.3 mm of clearance at each side, so check it before the head. Then the head and the back panel: steps 2 to 4 need them. Then the plate, the clips, the pod, the fan plate, and the grille.

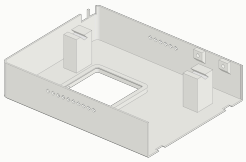
Print

The STL files are in their print orientation. Load them as they are.



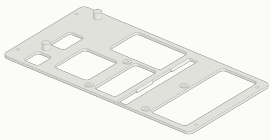
head

Navy · support check 0 g
Face down. The screen opening is in the first layers: use a smooth sheet for a clean face.



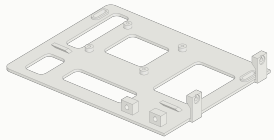
pod

Navy · support check 0 g
Outer face down. The grille pocket and the vents are in the first layers.



back

Orange · support check 0 g
Rear face down. A flat plate with the clip slots and the windows.



plate

Orange · support check 0 g
The face that goes on the cover down. The bosses and the hooks point up.



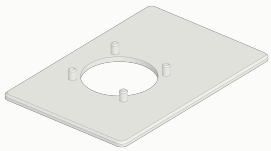
clip

Orange · support check 0 g
On its end, with a brim: the section of the clip is on the bed. Print 2.



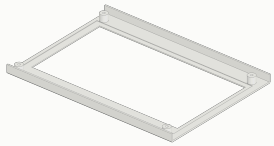
grille

White · support check 0 g
Back down. The snap tabs bend along the layers. The flat grille of the first pod: if you have it, do not print it again.



fan plate

White · support check 0 g
Grille side down. The four posts for the fan point up.



fit display

Navy · support check 0 g
Optional fit test of the screen opening. Face down.



roll nut

Orange · support check 0 g
Option until you have the metal T-nuts. On its end, with a brim. Print 2.

Settings (LulzBot TAZ Workhorse)

Profiles	One Cura profile for each part: run `python3 cura/make_profiles.py --install` with Cura closed, then pick “Printdeck - LulzBot …”.
Printer	LulzBot TAZ Workhorse, SE tool head, 0.5 mm nozzle
Material	PLA, 2.85 mm
Head, pod, grille, fan plate	0.3 mm layers, 1.5 mm walls, 1.2 mm top and bottom, 15 % grid infill, skirt
Back panel, plate	0.4 mm layers (a 0.3 mm first layer), 1.5 mm walls, 15 % grid infill, skirt
Clip, roll-in nut	0.3 mm layers, 100 % infill, with a brim.
Speeds	Infill 75 mm/s (60 mm/s with 0.4 mm layers), inner walls 60 mm/s, outer walls 35 mm/s (30 mm/s on a face that shows), travel 200 mm/s. The flow stays at 12 mm³/s or less.

Hardware

6×	**M3 × 10 socket cap head** 2 × plate on the cover (upper pair of PP-FP0164); 4 × back panel into the head · VIGRUE M3 kit
2×	**M3 × 12 socket cap head** 2 × plate flange into the back panel · VIGRUE M3 kit
12×	**M3 × 6 socket cap head** 4 × display board (it cuts its thread); 4 × Pi (it cuts its thread); 4 × drive on the back panel · VIGRUE M3 kit
4×	**M3 × 8 socket cap head** 2 × pod onto the plate; 2 × rail clip into its T-nut · VIGRUE M3 kit
2×	**M2.5 self-tapping round head Phillips, 8 to 11 mm long** 2 × fan onto the fan plate (it cuts its thread) · your fan screws
2×	**M3 hex nut, 5.5 mm across the flats** 2 × printed roll-in nuts, in place of the T-nuts (option) · M3 hex nuts
2×	**M3 roll-in T-nut with spring ball, for 2020 extrusion, 6 mm slot** 2 × rail clips · buy: uxcell M3 roll-in spring T-nut with ball, 2020 series (Amazon B09MFT1PKF)
2×	**M3 washer** 2 × under the cover screws · M3 washers
2×	**M3 × 5 heat-set insert** 2 × plate, pod screw blocks (3 to 5 mm fits) · FFVRVSS insert kit
6×	**M3 × 7 heat-set insert** 4 × head, back screws (3 to 7 mm fits); 2 × back panel, flange screws (3 to 7 mm fits) · FFVRVSS insert kit

From the model: each screw, washer, and insert is in the model at its place. All screws are socket cap head screws. A shorter or a longer insert is fine, up to the length in brackets. All the screws are M3 socket cap screws. The screws of the display and the Pi cut their own thread in the bosses. The position of the rear cover screw is to be confirmed (page 5).

Electronics and cables

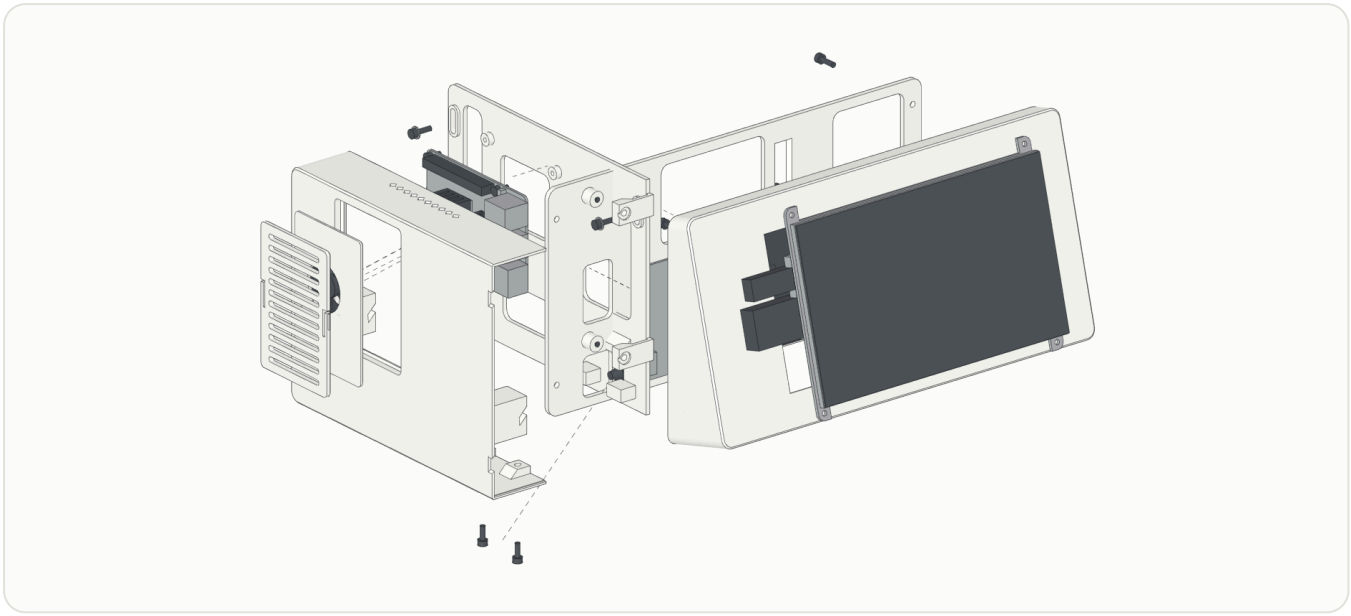
1×	Raspberry Pi 5 (from the Slate build)
1×	30 × 30 × 10 mm 5 V fan with the Pi 5 fan plug (4-pin JST-SH), the same as the Slate. From the fan to the fan socket of the Pi, the cable route is 4 cm. The cable must be 8 cm or more, so that the fan plate with the fan comes out of the pod on its cable.
1×	Stick-on heatsinks for the Pi 5 (a set), max. 13 mm high. The heatsink of the SoC can be up to 20 × 20 mm.
1×	Adafruit 7 inch HDMI display with touch, product 2407 (from the Slate build)
1×	2.5 inch SATA drive with its USB adapter board, the boot drive (from the Slate build)
1×	HDMI cable: micro HDMI to HDMI, straight plugs, a flat ribbon of 30 cm or more (the route is 27 cm). The ribbon kit of the Slate fits.
1×	Micro USB to USB-A cable, straight plugs, max. 3.5 mm thick, 15 cm or more (the route is 13 cm).
1×	USB 3 cable for the drive: micro-B SuperSpeed to USB-A, 15 to 20 cm (the route is 15 cm).
1×	USB-A to USB-B cable for the printer: 31 cm in the pod, plus the run behind the printer to the USB socket of the printer board.
1×	USB-C power cable: 9 cm in the pod, plus the run behind the printer to the Raspberry Pi 27 W USB-C power supply.

Tools

- Drill with a 3.2 mm bit (for the four holes of the Pi)
- Soldering iron with a heat-set insert tip, at 220 °C
- 2.5 mm hex key (M3 socket cap screws)

2 mm foam tape, one strip (between the panel and the display board).

On the printer



The mount uses two screws of the printer and the front rail. You drill no hole.

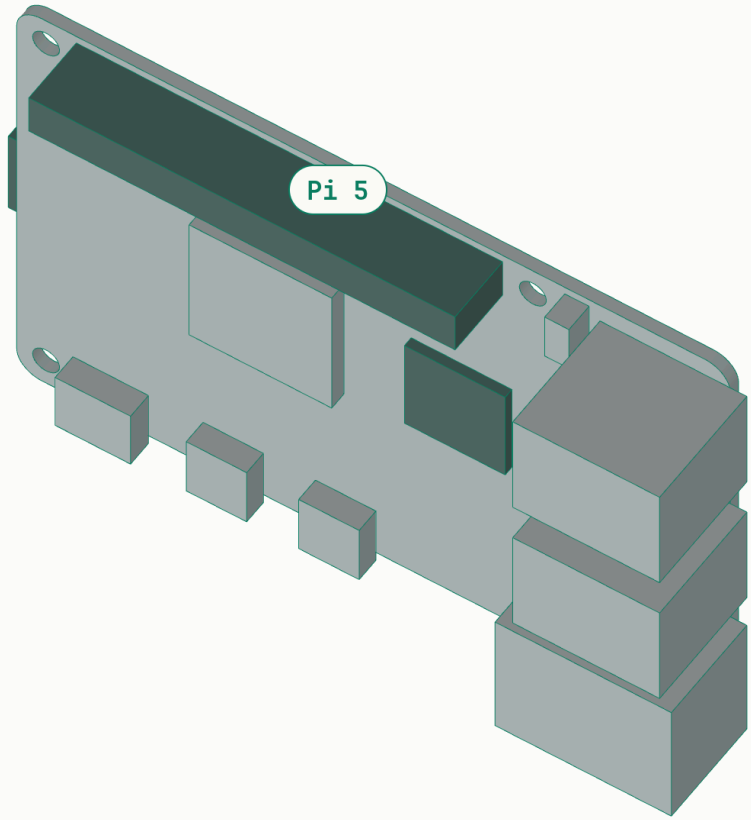
- **Cover screws.** The cover of the chassis is PP-FP0164, with 10 M3 screws into nuts in the chassis. The two screws of its upper pair, under the plate, change to M3 × 10 socket cap screws with a washer. They are at: 239.8 mm above the bottom of the chassis and 6.1 mm from its front face (front: measured (237.3 mm up, 3.6 mm from the cover edge)); 239.8 mm above the bottom of the chassis and 163.9 mm from its front face (rear: from the drawing, mirrored; to be confirmed). The pads have slots 10 mm long and 4.4 mm wide, so the plate fits 5 mm up or down.
- **Front rail.** Two printed clips hold on the front slot of the front left rail. An M3 roll-in T-nut for 2020 extrusion rolls into the slot behind the lips, and an M3 × 8 screw pulls it up. You drill no hole in the rail. The dovetail keys of the clips hold the right end of the head against the rail.
- **Flange.** Two M3 × 12 socket cap screws go from inside the pod through the flange of the plate into the back panel. They hold up the head.
- **Hooks.** The pod slides up onto two 45° hooks of the plate, and two M3 × 8 screws hold it from below.

Check on your printer

1. Check the rear screw of the upper pair on the cover: 237.3 mm above the bottom edge of the cover, and about 3.6 mm from the rear edge of the cover. The design uses the value of the LulzBot drawing.
2. Measure the TAZ Workhorse logo on the front of the chassis. The top of the head is 260 mm above the bottom of the chassis; the LulzBot drawing gives the logo at 264 mm, and the drawing is not to scale.
3. Look at the front slot of the front left rail. The clips need a free slot from 150 to 230 mm above the bottom of the chassis: remove the dust cover there, if the rail has one.
4. Move the bed to the front and the X axis to the bottom. Check that the head is clear of the bed, the print, and the tool head: it is 72 to 324 mm to the right of the left face of the chassis, and 125 to 260 mm above the bottom of the chassis.
5. Check behind the printer: the printer USB cable and the power cable leave the pod at the rear face of the chassis, 128 to 149 mm above its bottom.

1 Drill the Pi

`Pi 5` `3.2 mm drill`

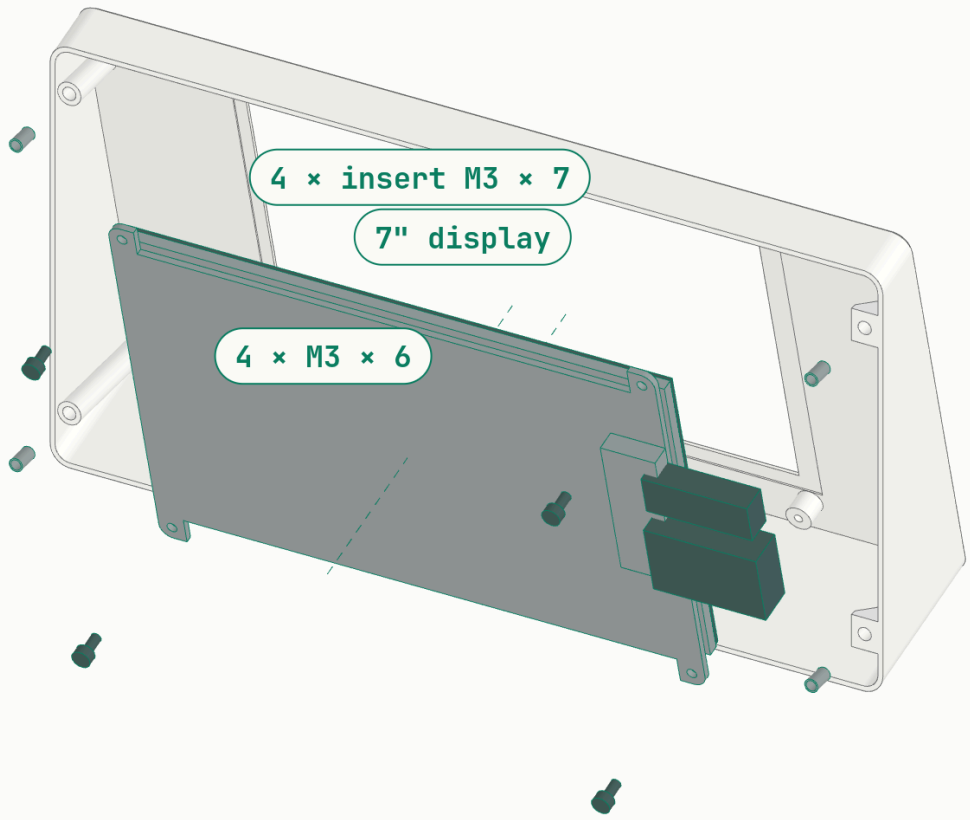


1. The mounting holes of the Pi 5 are 2.7 mm, for M2.5. Drill the four holes to 3.2 mm for M3.
2. Drill slowly, from the top of the board, with the board flat on a piece of wood.
3. Keep the drill straight. Stay inside the plated ring of each hole: do not touch the tracks or the parts around it.
4. Remove all the dust from the board with a brush or with air before you continue.

Note. Skip this step if your Pi already has 3.2 mm holes.

2 Display into the head

head · display · foam tape · 4 × M3 × 7 insert · 4 × M3 × 6

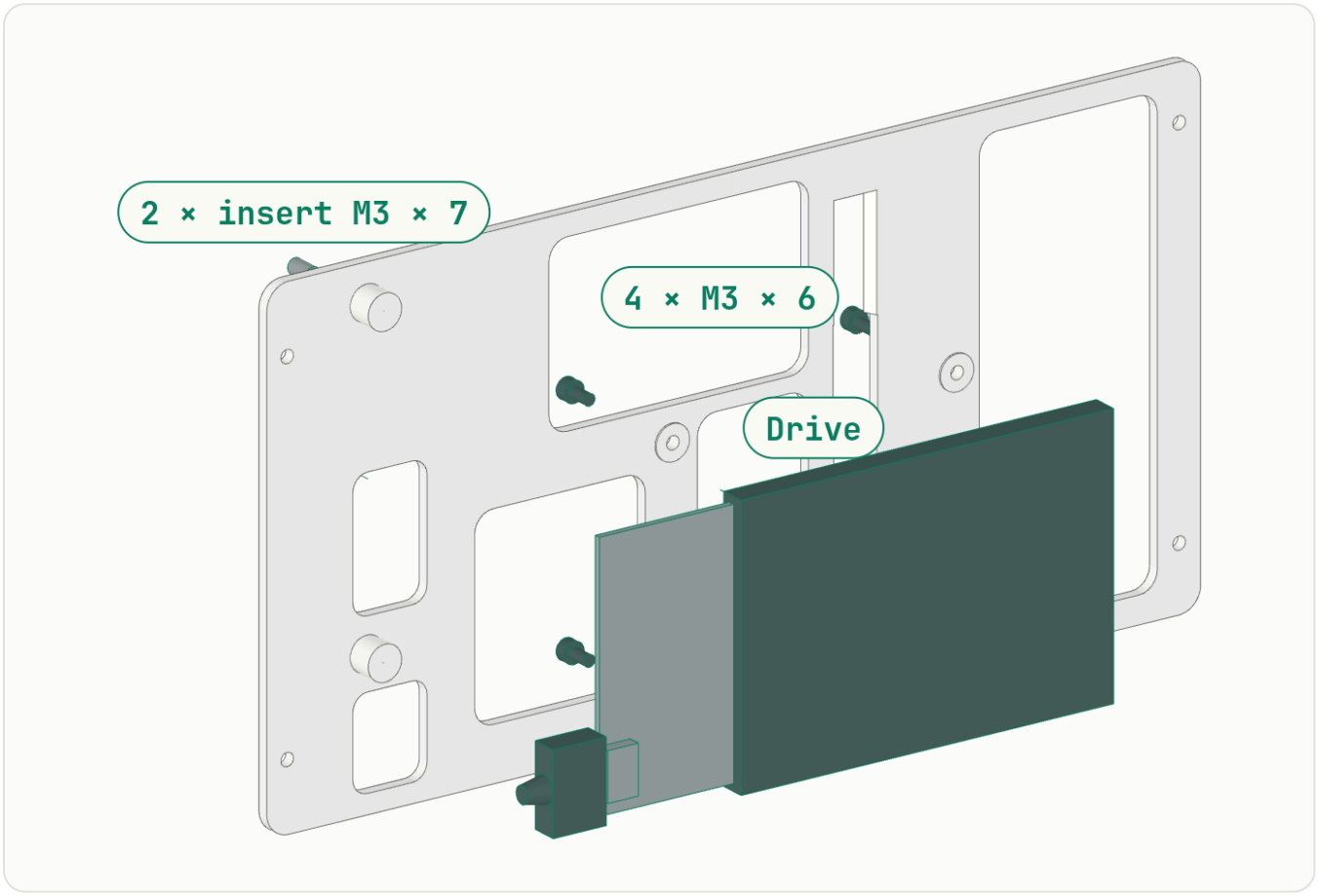


1. Press the four M3 × 7 inserts into the two columns and the two blocks at the back of the head. Use a soldering iron at 220 °C. Stop when the insert is flush.
2. Put the head face down on a soft cloth.
3. Put the display glass down into the pocket: the sockets of the board at the left (seen from the front), the FPC edge of the panel at the top.
4. Put the foam tape on the back of the panel. Put the board on the four bosses, and attach it with four M3 × 6 screws. They cut their own thread in the bosses: turn them until they are snug, and no more.
5. Connect the HDMI cable and the micro USB cable to the board. Both plugs point to the left.

Note. The screen opening is at the centre of the face: the border is the same at the left and the right.

3 Drive on the back panel

back panel | drive | 2 × M3 × 7 insert | 4 × M3 × 6

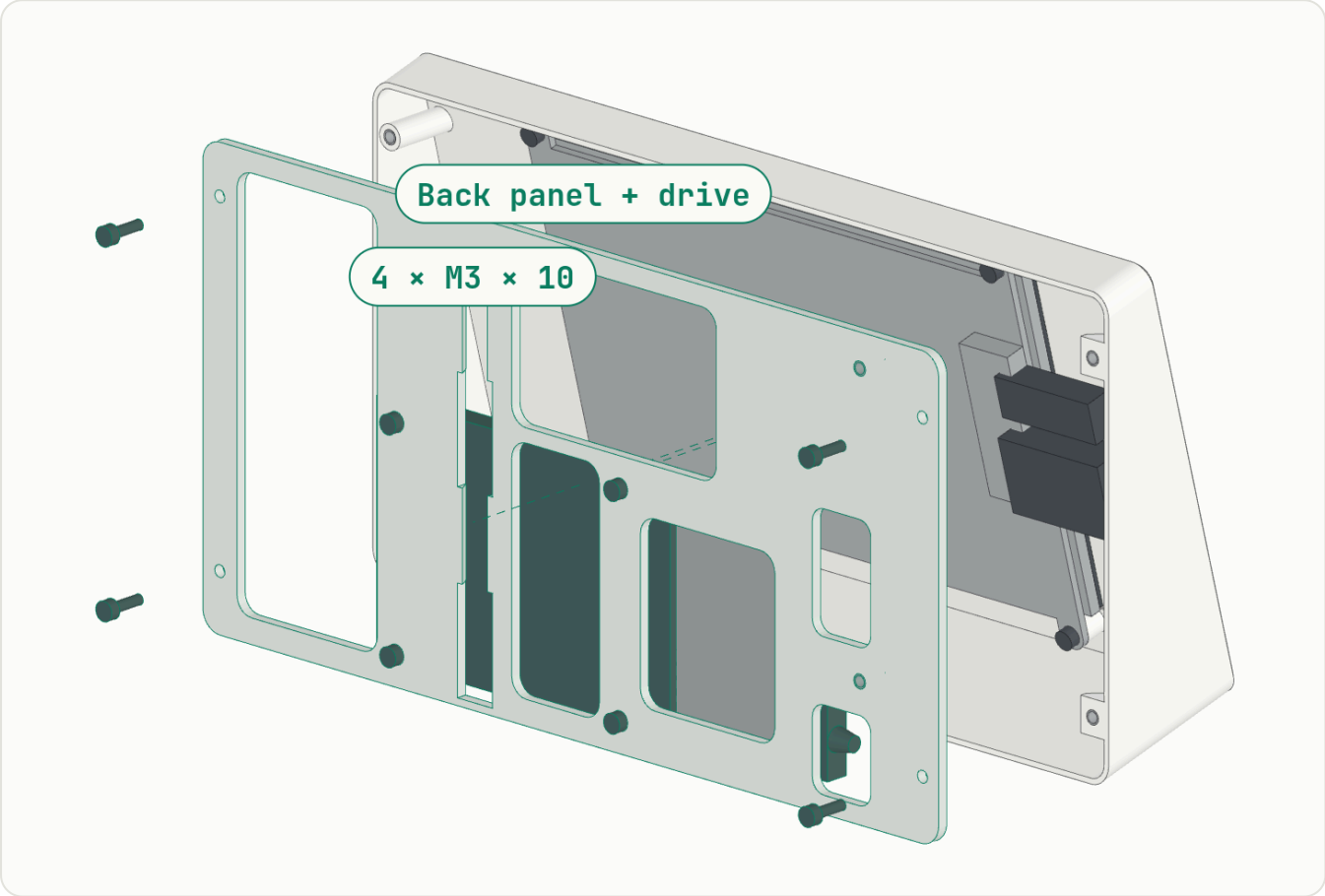


1. Press the two M3 × 7 inserts into the two bosses on the front face of the back panel, from the rear.
2. Put the drive on the four pads on the front face of the back panel: the adapter toward the left (the pod side), the M3 holes on the pads.
3. Attach it with four M3 × 6 screws from the rear face of the panel.
4. Connect the micro-B plug of the drive cable to the adapter.

Note. Press each insert in until it is flush with the rear face, or a little below it. Cut off any ring of melted plastic around the insert. The plate flange has a relief for a small ring, but a large one holds the flange off the back panel, and the two screws then bend the panel at its corner.

4 Back panel onto the head

head (step 2) back panel (step 3) 4 × M3 × 10

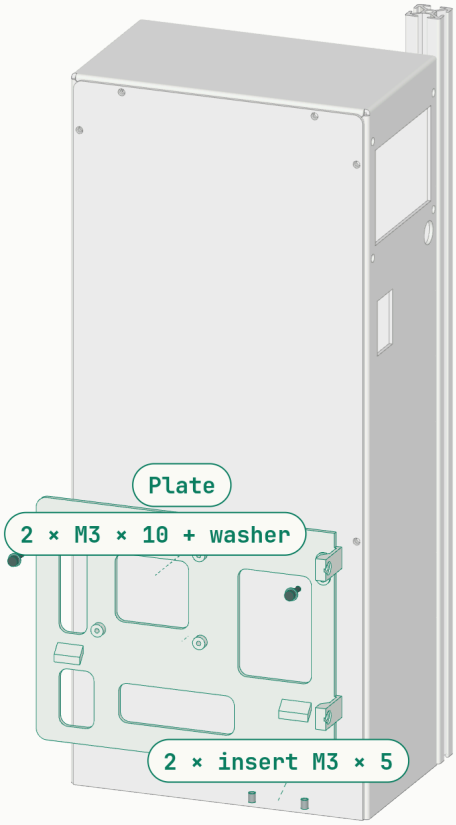


1. Put the display cables through the upper window at the left of the back panel, and the drive cable through the lower window.
2. Put the back panel on the head. Attach it with four M3 × 10 screws from the rear face, into the inserts.

Note. The heads of the screws stay on the rear face. They are behind the head, and they do not show.

5 Plate on the cover

plate | 2 × M3 × 5 insert | 2 × M3 × 10 | 2 × M3 washer

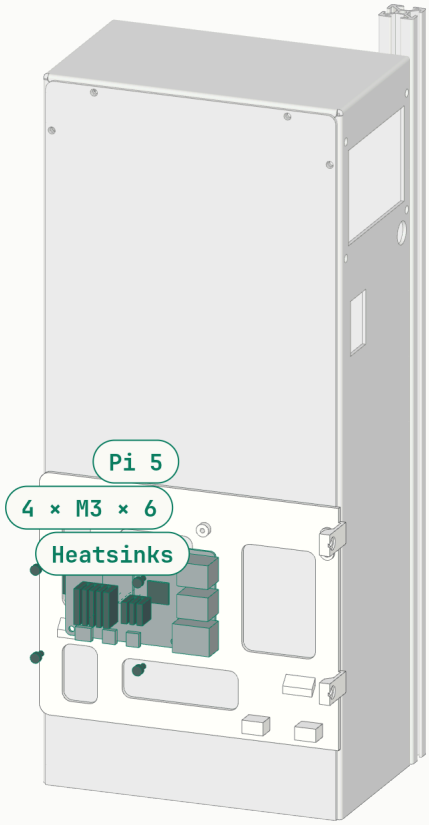


1. Press the two M3 × 5 inserts into the two blocks at the bottom of the plate (from below).
2. The cover is PP-FP0164, with 10 screws. Remove its upper pair of screws from the left side of the chassis: the front one and the rear one, 237.3 mm above the bottom edge of the cover.
3. Put the plate on the cover, with the slots of the two pads over the two holes. Attach it with two M3 × 10 screws, each with a washer.

Note. The slots in the pads let the plate fit 5 mm up or down and 0.5 mm to the front or the rear. The pad is 4 mm thick, so the M3 × 10 goes 3.5 mm into the nut of the chassis (the old screw went 4.0 mm).

6 Pi on the plate

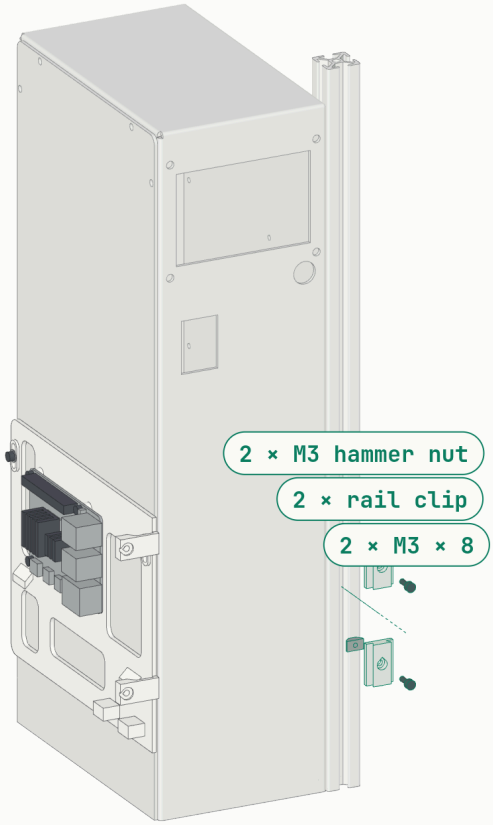
Pi 5 (step 1) heatsinks 4 × M3 × 6



1. Stick the heatsinks on the Pi: on the SoC (the large chip at the middle) and on the RP1 (the chip between the SoC and the USB ports). A heatsink can be up to 13 mm high.
2. Put the Pi on the four bosses: the USB ports toward the front of the printer, the USB-C and HDMI ports down.
3. Attach it with four M3 × 6 screws. They cut their own thread in the bosses: turn them until they are snug, and no more.

7 Clips in the rail

2 × rail clip | 2 × M3 roll-in T-nut | 2 × M3 × 8

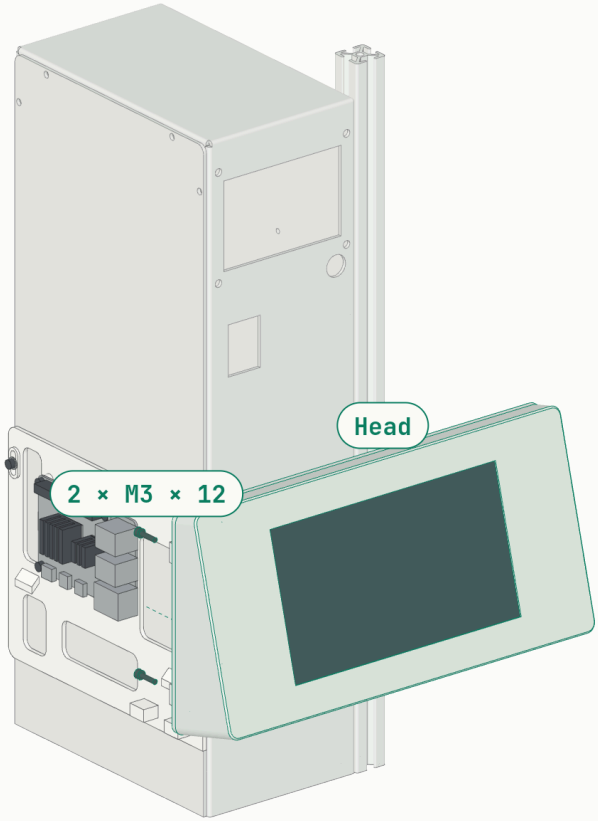


1. Hold the back panel on the front face of the front left rail, with its bottom edge 125 mm above the bottom of the chassis. Mark the top of the two long slots in the panel on the rail.
2. At each mark, put one edge of a T-nut into the front slot of the rail, with the thread toward you. Roll the nut in until it is behind the lips. The spring ball holds it in the slot.
3. Put a clip on the rail at a mark, with its key along the rail. Put an M3 × 8 screw through the counterbore in the key, and turn it into the nut.
4. Tighten the screw until the clip is tight on the rail. Do the same with the second clip.

Note. You can roll the nut into the slot at any point on the rail. You do not remove the ends of the rail. If the nut moves away from the mark, push it back with a small screwdriver. Until you have the metal T-nuts, you can use the printed roll-in nuts (stl/roll_nut.stl). Press an M3 hex nut into the pocket of each one with pliers. Put one long edge into the slot, with the hex nut away from you, and roll the nut in behind the lips. Tighten the screw gently: the printed nut is not as strong as a metal T-nut.

8 Hang the head

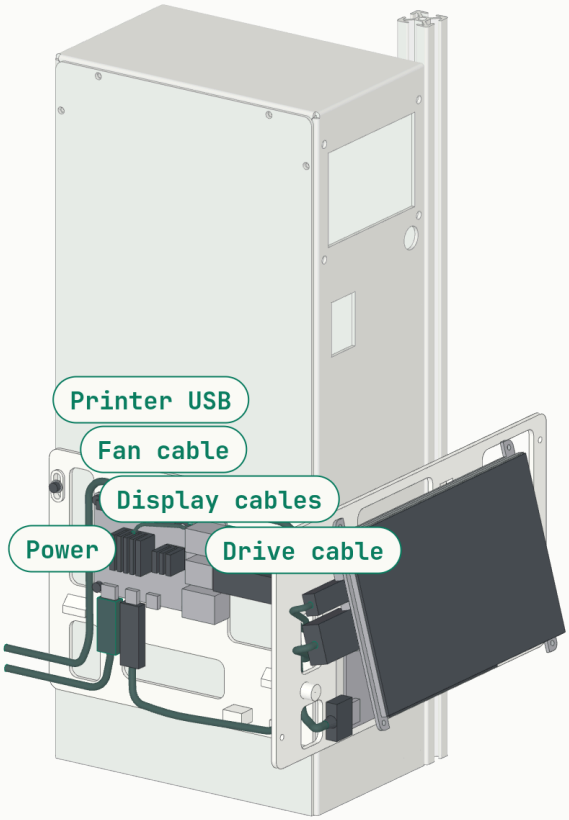
`head (step 4)` `2 × M3 × 12`



1. Hold the head 30 mm above its place, with the two windows of the back panel over the two clip keys. Push the back panel against the clips.
2. Let the head drop 30 mm. The keys go into the dovetail slots.
3. Lead the display cables and the drive cable out of the windows into the pod.
4. From inside the pod, put the two M3 × 12 screws through the flange of the plate into the back panel.

Note. The two screws through the flange hold up the head. The clips hold its right end against the rail.

9 Cables

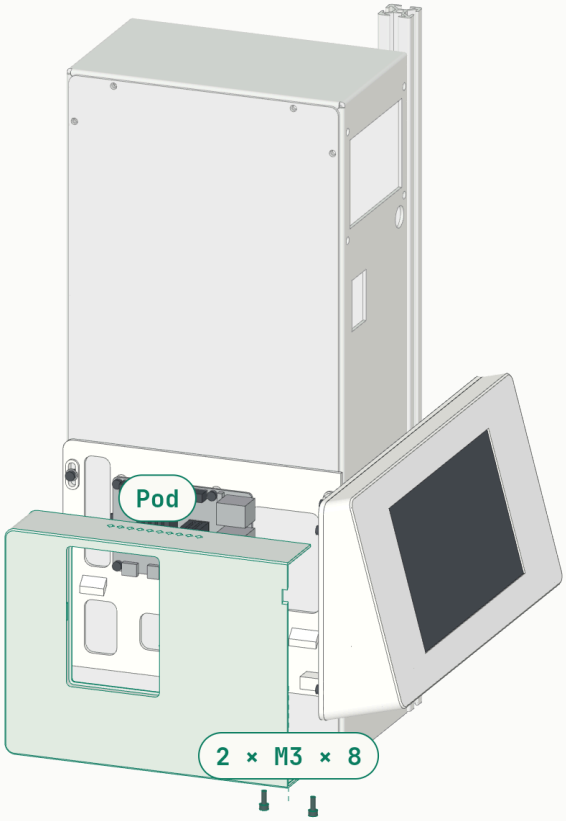


1. Connect the HDMI cable to the HDMI 0 port (next to the USB-C port). Push the plug straight in. Let the cable go straight down from the plug, then turn it to the front in a wide curve along the bottom of the pod. Do not bend it at the plug.
2. Connect the touch cable to the upper USB 3 port (blue), and the drive cable to the lower USB 3 port.
3. Connect the printer cable to the lower USB 2 port. Lead it up the front of the pod, over the Pi, down at the rear end of the Pi, and out of the upper notch in the rear wall.
4. Connect the power cable to the USB-C port. Lead it down and to the rear, out of the lower notch.
5. Behind the printer, connect the printer cable to the USB socket of the printer board, and the power cable to the power supply.

Note. No cable goes through a hole in the printer, and no cable shows on the front or the side.

10 Pod

pod · 2 × M3 × 8



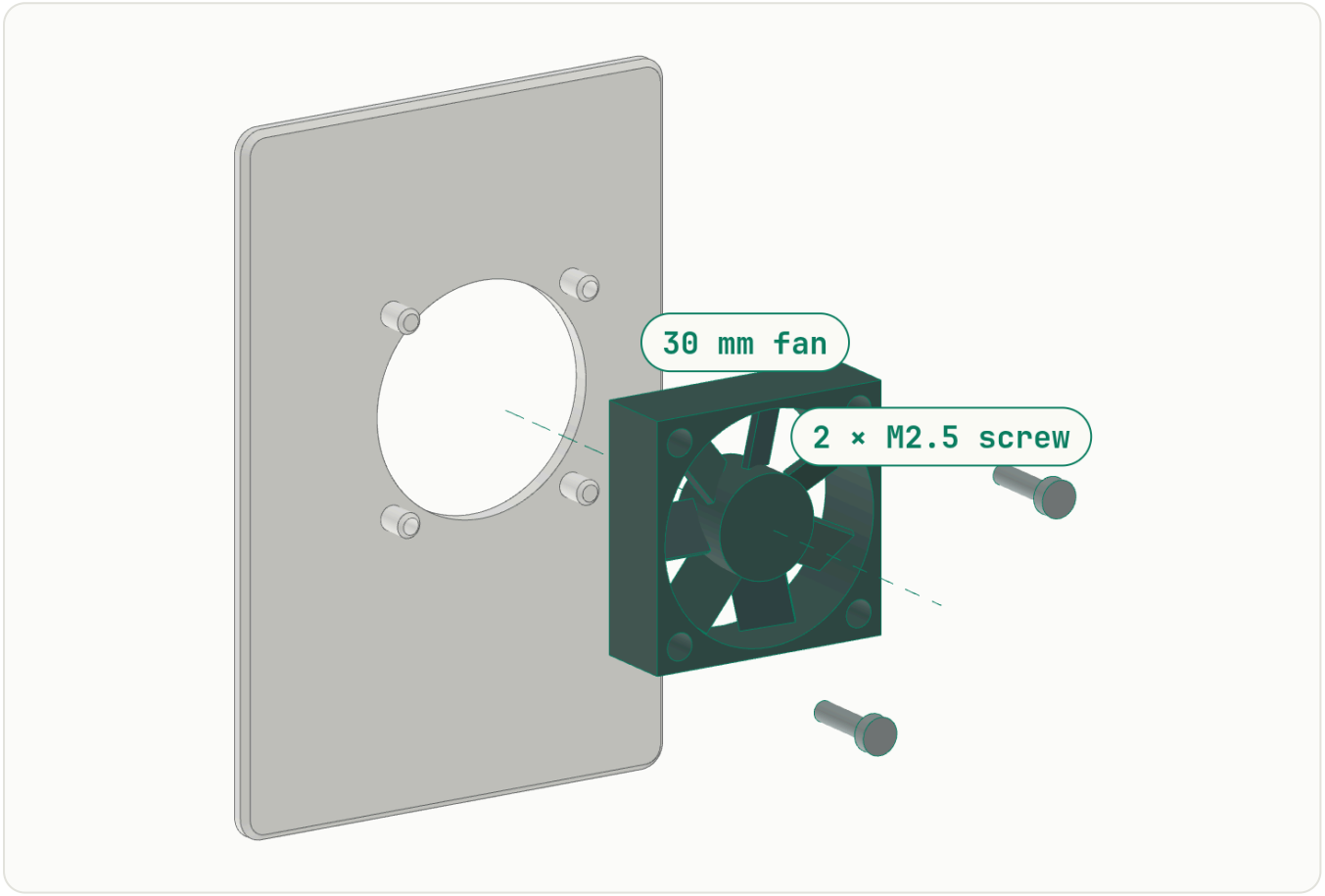
1. Connect the fan cable to the fan socket of the Pi: the small 4-pin socket between the end of the GPIO header and the USB ports. Lay the fan on the Pi.
2. Hold the pod about 6 mm low and 6 mm out from its place, and 4 mm to the rear. Move it 4 mm to the front, then push it up and in at 45° onto the two hooks of the plate. The rear wall goes over the two cables.
3. Put the two M3 × 8 screws up through the bottom of the pod into the plate. They pull the pod up, and the 45° hooks pull it onto the plate.

Note. Two notches in the front edge of the pod are for the heads of the two screws into the left wall of the head.

11 Fan onto the fan plate

`fan plate` `fan`

`2 × M2.5 self-tapping round head Phillips, 8 to 11 mm long`

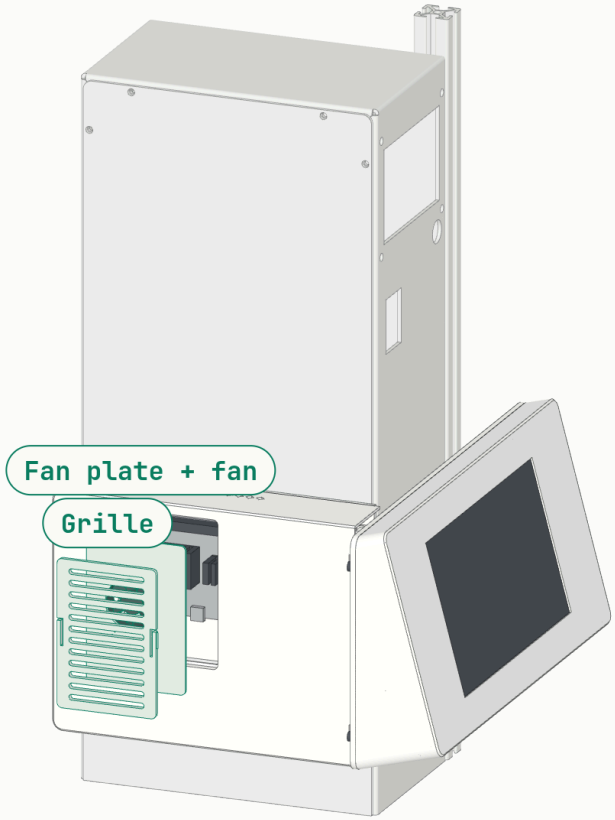


1. Take the fan out of the pod through the grille pocket. Its cable stays on the Pi.
2. Hold the fan plate with its four posts toward you. Turn the fan so that it blows away from the fan plate, onto the Pi. Most fans show the direction of the air with an arrow on the frame.
3. Push the fan onto the four posts, until it is flat on the fan plate. The two solid posts hold it in place.
4. Put the two M2.5 screws into the other two holes, from the far face of the fan. Turn them into the hollow posts until they are snug, and no more. They cut their own thread.

Note. A screw of 8 to 11 mm fits: it goes through the fan hole into the post. A 10 mm screw has 4 mm of thread. The fan cable leaves the frame from a side: it does not touch the posts or the screw heads.

12 Fan plate and grille

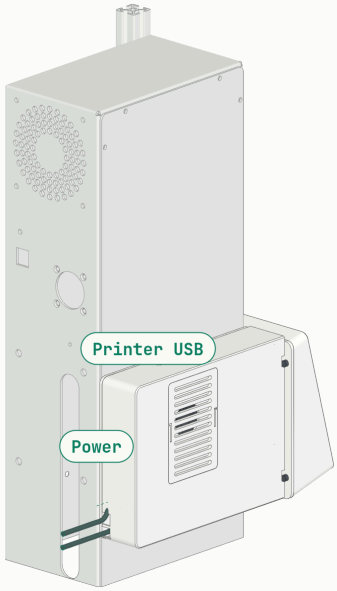
`fan plate with the fan (step 11)` `grille`



1. Push the fan plate with the fan into the grille pocket of the pod from the outside, the fan first. The fan plate goes down onto the ledge of the pocket. The cable stays inside the pod.
2. Hold the grille at the pocket, with its tabs at the sides. Push it into the pocket over the fan plate. Push both sides until the two tabs click. The grille is flush with the pod.

Note. To get to the fan, put a small screwdriver into a vent slot near a tab, and lift the grille out. Then pull the fan plate out at its air hole. The fan comes out with it, on its cable.

Connect



Cable	Length	Route
HDMI	27 cm	HDMI 0 port to the display, plug to plug
Touch USB	13 cm	upper USB 3 port to the display, plug to plug
Drive	15 cm	lower USB 3 port to the drive, plug to plug
Printer USB	31 cm	lower USB 2 port to the rear face of the chassis
Power	9 cm	USB-C port to the rear face of the chassis

Wide display. The display is turned 180° in the head (its sockets are at the left). Before you connect the drive, put it in a computer and open `printdeck.txt` on the PRINTDECK partition. Set `display_rotate=180`: the deck then uses its wide layout. If the picture is upside down at the first start, set `display_rotate=0`.

1. Connect the power cable to the power supply behind the printer.
2. Connect the printer cable to the USB socket of the printer board.
3. Turn on the printer. The deck starts from the drive.

To open the mount

1. Lift the grille out with a small screwdriver in a vent slot, pull the fan plate out at its air hole, and disconnect the fan cable. Remove the two screws under the pod. Move the pod down and out at 45°, off the hooks, then 4 mm to the rear and out.
2. Disconnect the cables from the Pi. Remove the two flange screws from inside the pod.
3. Lift the head 30 mm and pull it off the clip keys. The display and the drive come off with it.
4. To get to the display or the drive, remove the four screws from the rear of the back panel.